

In the name of God



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Artificial Intelligence

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CA 4

Questions

1. توابع فعال‌ساز به چه منظور استفاده می‌شوند؟ دو مورد از آنها را نام ببرید و نحوه عملکرد آنها را بیان کنید.
2. نقش تابع هزینه در آموزش مدل‌ها چیست؟ دو مورد از آنها را نام ببرید و نحوه عملکرد آنها را بیان کنید. هر یک از این توابع در چه شرایطی مورد استفاده قرار می‌گیرند؟
3. Batch Normalization چیست و به چه منظور استفاده می‌شود؟
4. چه زمانی شبکه‌های عصبی نسبت به الگوریتم‌های یادگیری ماشین کلاسیک برتری دارند؟
5. اگر بخواهیم یک شبکه عصبی با ۱۰۰۰ پارامتر را تنها با ۱۰ داده آموزش دهیم، چه اتفاقی خواهد افتاد؟
6. خلاصه‌ای از چگونگی عملکرد SGD و Momentum ارائه دهید.
7. نقش لایه‌های پیچشی در این شبکه‌ها چیست؟ چرا پس از آنها لایه‌های تماماً متصل قرار می‌گیرد؟
8. توضیح دهید چرا برای استخراج ویژگی از لایه‌های تماماً متصل استفاده نمی‌شود؟

1. What are activation functions used for? Name and explain two of them.

Purpose: Activation functions introduce non-linearity to the network, allowing it to learn complex patterns and relationships in the data. Without activation functions, the model would essentially behave as a linear function, regardless of the number of layers.

Examples:

1. ReLU (Rectified Linear Unit):

- Formula: $f(x) = \max(0, x)$
- Behavior: Outputs the input directly if it's positive; otherwise, it outputs zero.
- Advantage: It helps with gradient flow and is computationally efficient.
- Limitation: Can suffer from the "dying ReLU" problem, where neurons can get stuck during training.

2. Sigmoid:

- Formula: $f(x) = \frac{1}{1+e^{-x}}$
 - Behavior: Maps input to a range between 0 and 1.
 - Advantage: Useful for probabilistic outputs.
 - Limitation: Can lead to vanishing gradients for large positive or negative inputs.
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2. What is the role of loss functions in training models? Name and explain two of them.

Purpose: Loss functions measure the difference between the predicted output and the true target, guiding the optimization process to minimize this difference.

Examples:

1. Mean Squared Error (MSE):

- Formula: $MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$
- Usage: Commonly used in regression tasks.
- Behavior: Penalizes larger errors more heavily due to squaring.
- Limitation: Sensitive to outliers.

2. Cross-Entropy Loss:

- Formula: $L = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c})$
- Usage: Used for classification tasks.
- Behavior: Measures the difference between the predicted probability distribution and the true labels.
- Advantage: Works well for probabilistic outputs, especially with softmax activation.

3. What is Batch Normalization, and why is it used?

Definition: Batch Normalization normalizes the inputs of each layer across a mini-batch to have a mean of zero and a variance of one.

Purpose:

- Improves training speed and stability by reducing internal covariate shift.
 - Allows for higher learning rates, reducing the chance of getting stuck in poor local minima.
 - Acts as a regularizer, reducing the need for other techniques like dropout.
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4. When do neural networks outperform classical machine learning algorithms?

Scenarios:

1. **High-dimensional data:** Neural networks excel when the input data has a very large feature space (e.g., images, text, and audio).
 2. **Large datasets:** Neural networks perform better when there's a lot of labeled data for training.
 3. **Complex patterns:** When the relationship between inputs and outputs is highly non-linear or hierarchical, neural networks are superior.
 4. **Transfer learning:** Pretrained neural networks can generalize well to new tasks, unlike many classical models.
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5. What happens if we train a neural network with 1,000 parameters on only 10 data points?

- **Overfitting:** The model will memorize the training data instead of learning generalizable patterns.
- **Poor generalization:** The model will perform well on training data but fail on unseen data.

- **Regularization won't help:** Even with techniques like L2 regularization or dropout, the small data size is insufficient.

6. Summary of how SGD and Momentum work

- **SGD (Stochastic Gradient Descent):**
 - Updates model parameters by computing the gradient for each mini-batch.
 - Formula: $\theta = \theta - \eta \nabla L(\theta)$
 - Advantage: Efficient for large datasets.
 - Limitation: Can oscillate and converge slowly.
 - **Momentum:**
 - Builds upon SGD by adding a fraction of the previous update to the current update.
 - Formula: $v = \gamma v - \eta \nabla L(\theta), \theta = \theta + v$
 - Behavior: Accelerates convergence in the direction of consistent gradients and reduces oscillations.
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7. What is the role of convolutional layers, and why are fully connected layers used afterward?

- **Convolutional layers:**
 - Extract spatial features like edges, textures, and patterns from the input.
 - Reduce the parameter count compared to fully connected layers for high-dimensional inputs.
- **Fully connected layers:**
 - Aggregate the extracted features to make final predictions.
 - Perform the classification or regression task based on the high-level feature representations.

8. Why are fully connected layers not used for feature extraction?

- Reason:
 - Fully connected layers treat all inputs equally and lose spatial relationships in the data.
 - They have a large number of parameters, making them inefficient for feature extraction, especially for high-dimensional data like images.
- Alternative:
 - Convolutional layers are used for feature extraction as they preserve spatial hierarchies and reduce the parameter count.